

What is Agentic AI? | Agentic AI using LangGraph

1. Introduction

- **Generative AI** mainly generates content such as text, images, or code.
- **Agentic AI** goes further by allowing AI systems to **make decisions and take actions** to achieve a goal.
- Agentic AI can use **memory, tools, reasoning, and planning**.

2. What is Agentic AI?

- Agentic AI is an AI system that can **work toward a goal with some level of autonomy**.
- Instead of only responding to a prompt, an agent can:
 1. Understand the goal.
 2. Decide what to do.
 3. Use available tools.
 4. Observe the result.
 5. Decide the next step.
- **Simple idea:** Generative AI generates an answer; Agentic AI can **decide and act**.

Example of Agentic AI

Imagine asking for an AI:

"Plan a trip to Islamabad within my budget."

A normal Generative AI system may simply provide a suggested itinerary.

An Agentic AI system could:

- Search for hotels and transportation.
- Compare available options.
- Check prices.
- Select options according to your budget.
- Create an itinerary.
- Adjust the plan if something is unavailable.

The important point is that the AI **performs multiple steps to achieve the goal**, rather than producing only one response.

Key Characteristics

The main characteristics of Agentic AI are:

- **Autonomy** — Can perform tasks with limited human intervention.
- **Goal-oriented behavior** — Works toward a specific objective.
- **Planning** — Breaks a complex goal into smaller steps.
- **Reasoning** — Decides what action should be taken.

- **Adaptability** — Changes its approach when circumstances change.
- **Context awareness** — Uses relevant information from the current situation.

3. Autonomy

- **Autonomy** means the AI can decide and perform actions without requiring a human to tell it every individual step.
- Example:
 - Human: "**Find the best flight within my budget.**"
 - Agent: Searches → compares → evaluates → selects the best option.
- More autonomy means **less step-by-step instruction from the human.**

4. Goal Oriented

- Agentic AI is driven by a **goal**, not simply by generating a response.
- The agent determines what actions are necessary to achieve that goal.
- Example:
 - Goal: "**Complete a research report.**"
 - Actions: Search information → analyze sources → organize findings → write report.
- The agent can decide the sequence of actions based on the goal.

5. Planning

- Planning means **deciding the steps required to accomplish a goal.**
- A complex task is divided into smaller tasks.
- Example:

Goal: Build a website

→ Understand requirements

→ Design pages

→ Write code

→ Test website

→ Fix errors

→ Deploy

- An agent can create and modify this plan as the task progresses.

6. Reasoning

- Reasoning allows an agent to **evaluate information and decide what to do next.**
- The agent considers:
 - Current situation
 - Available information
 - Possible actions

- Results of previous actions
- Example: If one tool fails, the agent can reason about whether to retry, use another tool, or change its approach.

7. Adaptability

- Adaptability means the agent can **change its behavior when new information or unexpected situations appear**.
- Example:
 - Planned hotel is unavailable.
 - Agent recognizes the problem.
 - Searches for alternatives.
 - Updates the plan.
- The agent does not always follow a fixed sequence.

8. Context Awareness

- Context awareness means understanding the **current situation and relevant information**.
 - An agent considers things such as:
 - Previous actions
 - User requirements
 - Current task state
 - Tool results
 - Available resources
 - This helps the agent make **more relevant decisions**.
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2. Components of Agentic AI

An Agentic AI system commonly contains several important components:

Brain + Orchestrator + Tools + Memory + Supervisor

Each component has a different responsibility.

1. Brain

- The **Brain** is usually the **LLM**.
- It handles:
 - Understanding instructions
 - Reasoning
 - Making decisions
 - Generating responses
- Examples include GPT, Claude, and other language models.

Simple idea: The brain decides **what should be done**.

2. Orchestrator

- The **Orchestrator** manages the agent's workflow.
- It decides:
 - Which step should run.
 - Which component should be called.
 - What happens after an action.
 - When the task is complete.
- **Simple idea:** The orchestrator controls **how the work flows**.

3. Tools

- Tools allow an agent to **interact with the outside world**.
- Examples:
 - Web search
 - Calculator
 - Database
 - APIs
 - Code execution
 - File systems
- Without tools, an agent may only reason and generate text.
- With tools, it can **perform real actions and access external information**.

4. Memory

- Memory allows an agent to **retain and use information**.
- It can help the agent remember:
 - Previous interactions
 - Important user information
 - Previous task results
 - Current task state
- Memory helps maintain **continuity across multiple steps or interactions**.

5. Supervisor

- A **Supervisor** manages or coordinates agents/components.
- It can decide **which agent should handle a particular task**.
- Example:

User request → Supervisor → Research Agent → Writing Agent → Final Result

- This becomes especially useful when a system contains **multiple specialized agents**.